DeepFold3 Kernel Benchmark Report

Recommended Optimizations

GPU: NVIDIA GeForce RTX 5090 | dtype: bfloat16 | 2026-03-16

Optimization Strategy

DeepFold3 Triton Kernel Optimization Stack
(adapted from NVIDIA cuEquivariance design patterns)

ATTENTION (104 modules, 3.7-4.6x speedup)
  1. Fused EvoformerAttention — Triton FlashAttention with pair bias
     • Autotuned tile sizes (BLOCK_Q/KV x num_stages x num_warps)
  2. Fused Pair Bias — LN + tl.dot GEMM(128→4) + permute + mask
     • 5-10x faster than old scalar loop; supports training backward
  3. Deferred Col Transpose — skip N^2C copy for column attention
     • Col overhead: 52% → 8% vs row attention

TRIANGLE MULTIPLY (96 modules, 1.2-1.3x speedup)
  Level 3: Fused Dual GEMM — sigmoid(x@W1+b1) * (x2@W2+b2) * mask
     • TWO_INPUTS mode, bias, mask fusion, TF32 precision control
     • Eliminates Incoming/Outgoing memory asymmetry (+1.5GB → 0)
     • Backward: pre-GEMM kernel + cuBLAS (split pattern from cuEQ)
  Level 2: Fused LN+Linear for post-einsum norm_center + out_proj
  Level 1: Fused LN+packed GEMM+sigmoid for pre-einsum gate
  • einsum replaced with torch.matmul on permuted views (zero-copy)

TRANSITIONS (context-sensitive)
  Pair transitions ($M=N^2$, C=128): Triton fused → 1.2-1.3x
  Single transitions (M=N, C=384): PyTorch baseline faster (M too small)

INFRASTRUCTURE
  • Precision enum: DEFAULT / TF32 / TF32x3 / IEEE
  • Persistent autotuning cache: ~/.cache/deepfold3-triton/
  • GPU fingerprinting with fuzzy matching (20% tolerance)

OptimConfig Presets

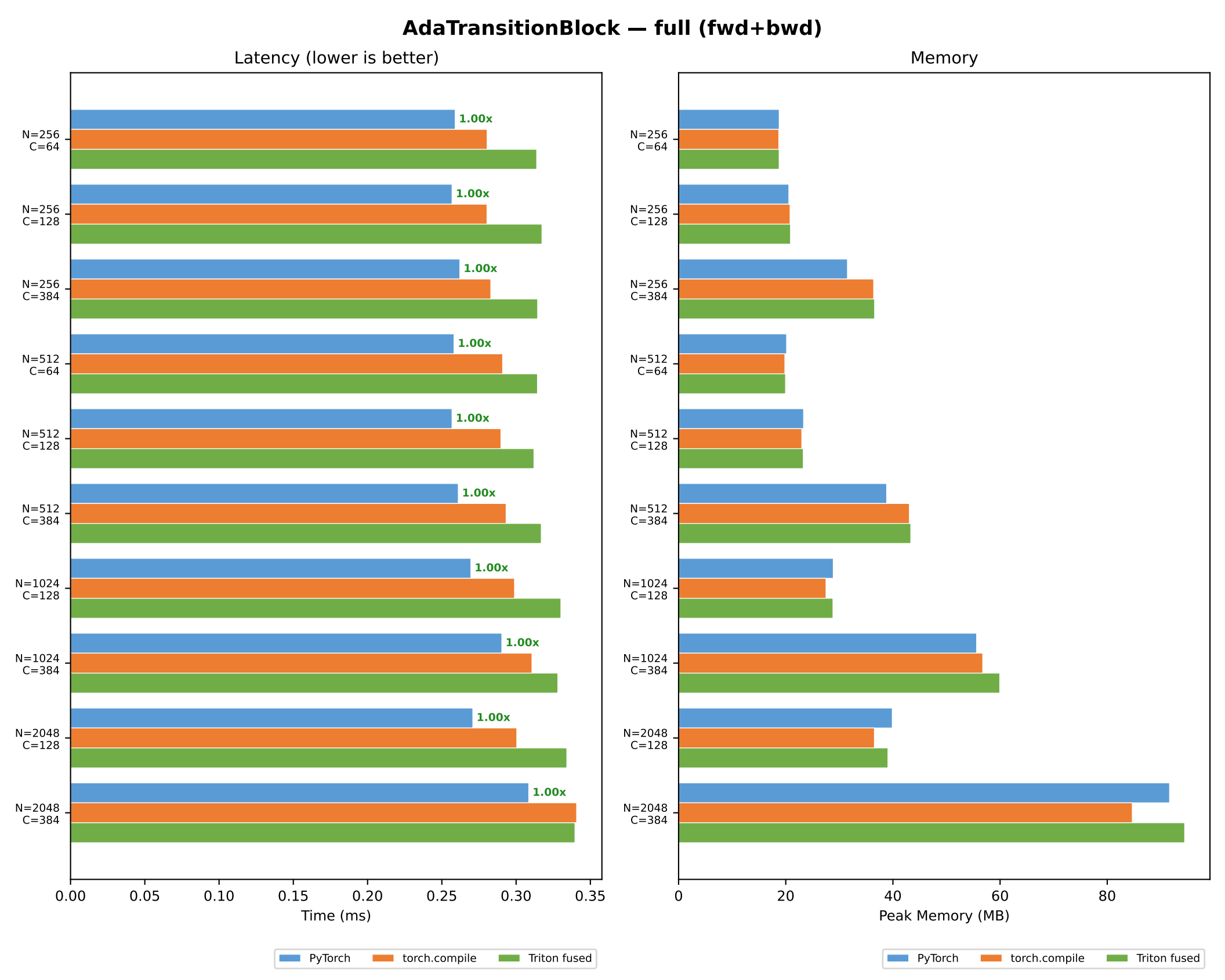
```
OptimConfig.recommended()  —  Best overall (training + inference)
  ✔ use_fused_evo_attention      2.9-3.3x attention speedup
  ✔ use_fused_pair_bias           +25-30% on top (training + inference)
  ✔ use_fused_triangle_mul        1.2-1.3x TriMul + fixes mem asymmetry
  ✔ use_fused_transition_pair      1.2-1.3x pair transitions (M=N²)
  ✔ use_torch_compile              1.3-2.3x for MSA/Single attention

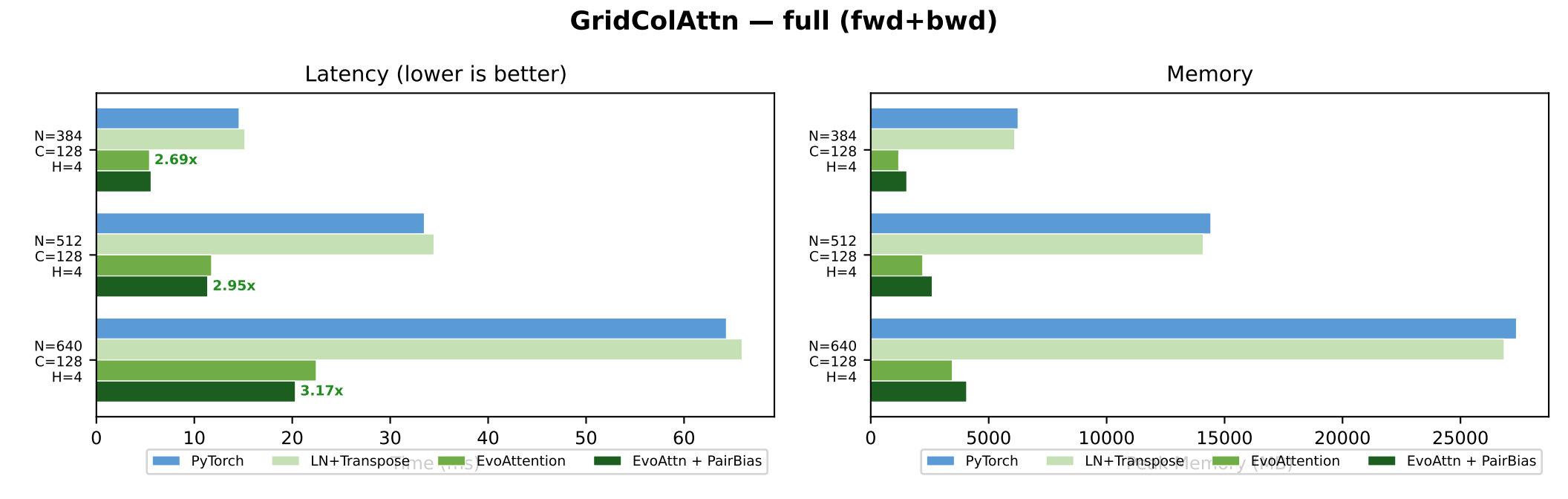
OptimConfig.inference()  —  Maximum inference speed
  ✔ Same as recommended, minus torch.compile
  ✔ use_fused_ln_transpose         1.7-2.3x LN+transpose (forward only)

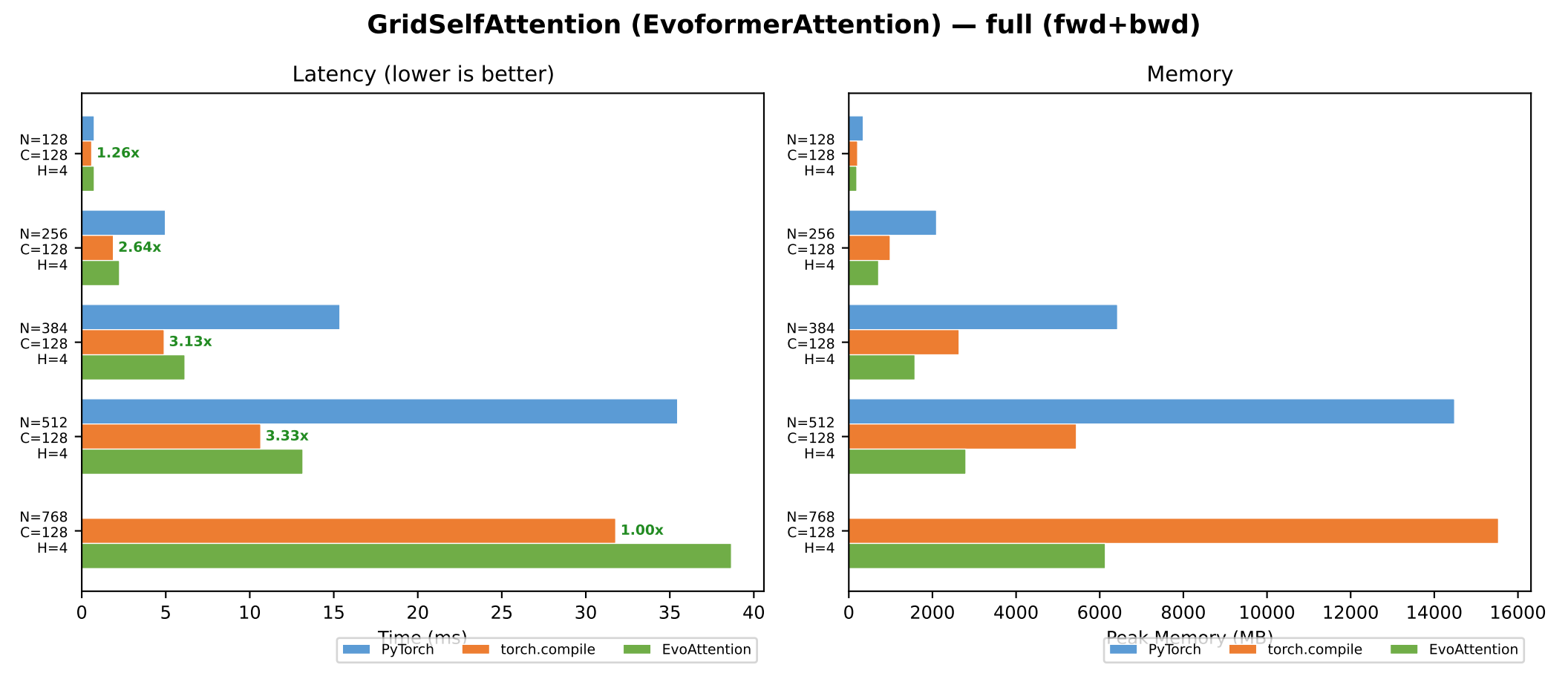
OptimConfig.triton_only()  —  All beneficial Triton kernels
  ✔ All fused kernels that demonstrably outperform PyTorch
  ✘ No torch.compile (avoids compilation overhead)

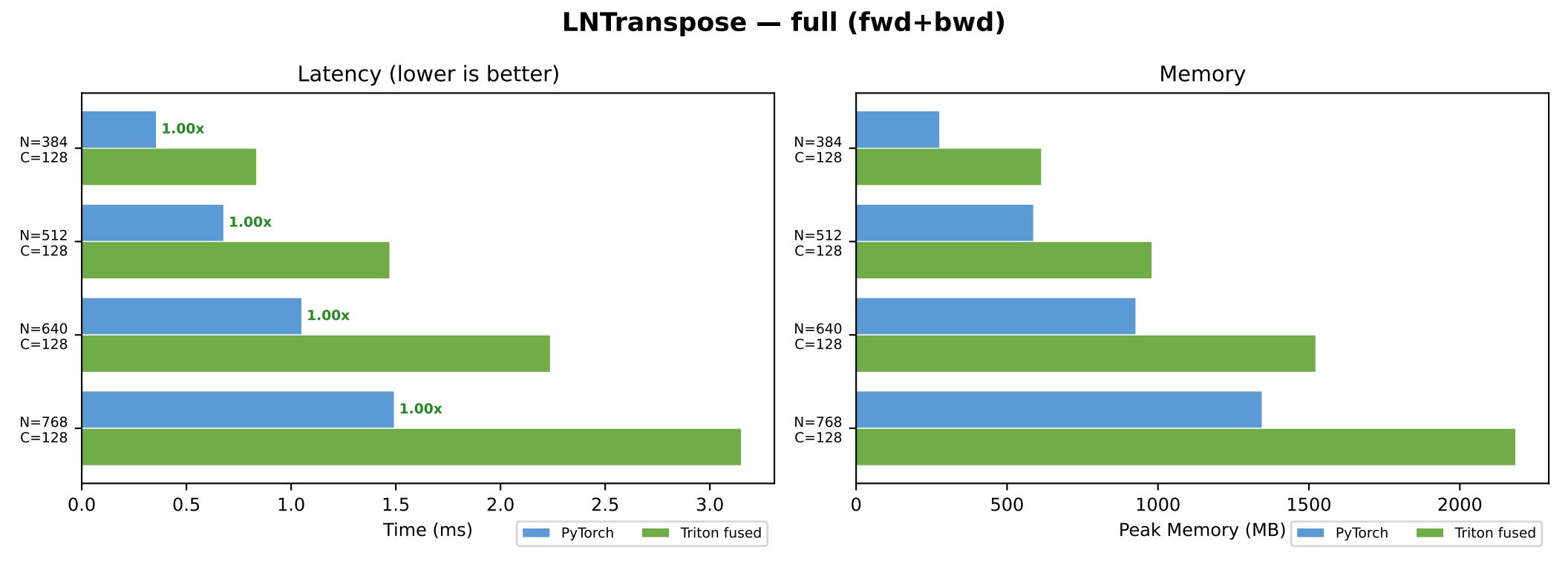
OptimConfig.all_enabled()  —  For testing only
  ✔ Everything including slow single/ada transitions

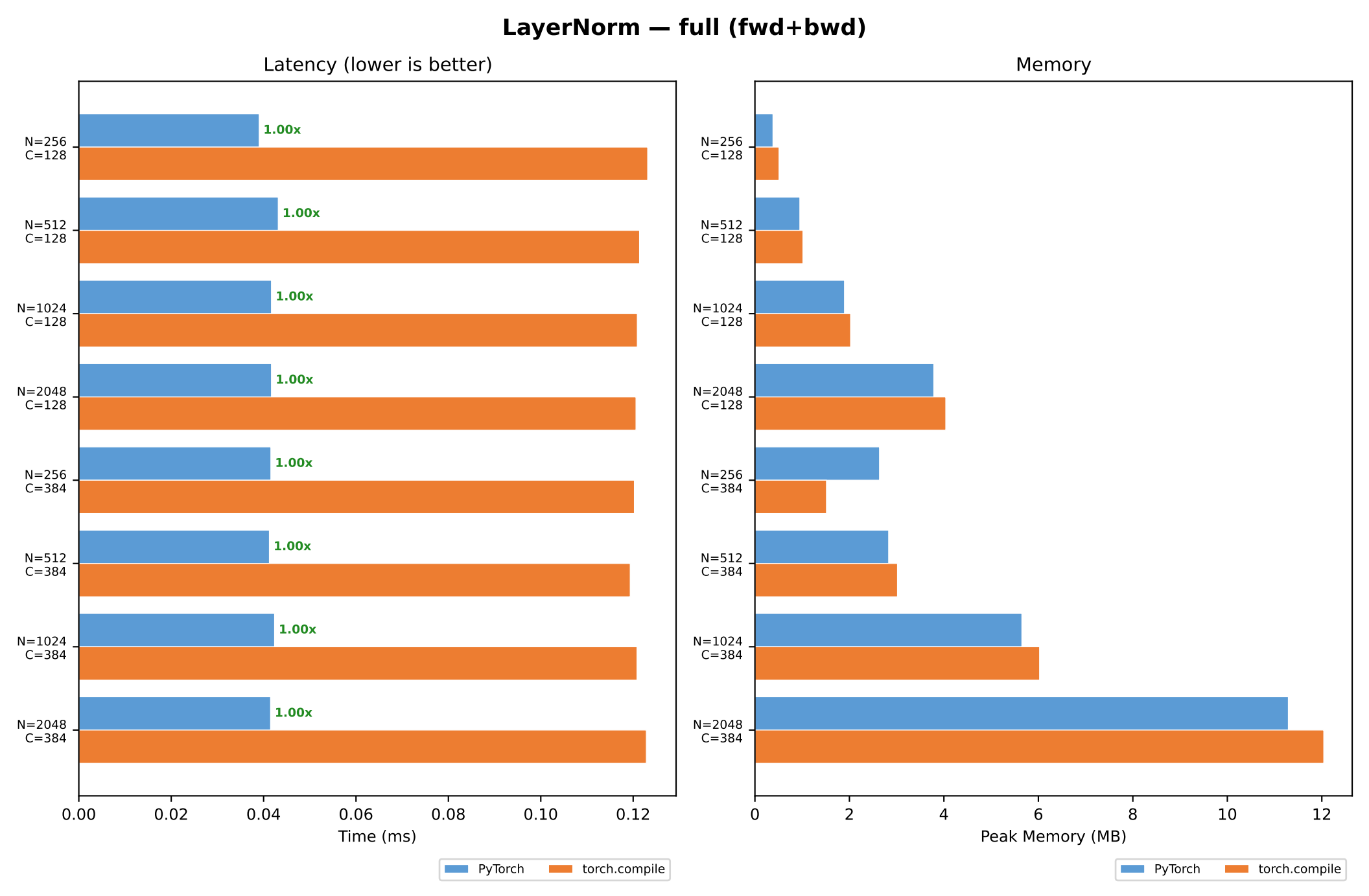
Flags NOT in recommended (benchmarked slower than PyTorch):
  ✘ use_fused_transition_single    0.8x at M=N, C=384
  ✘ use_fused_ada_transition       0.8x at M=N, C=768
```

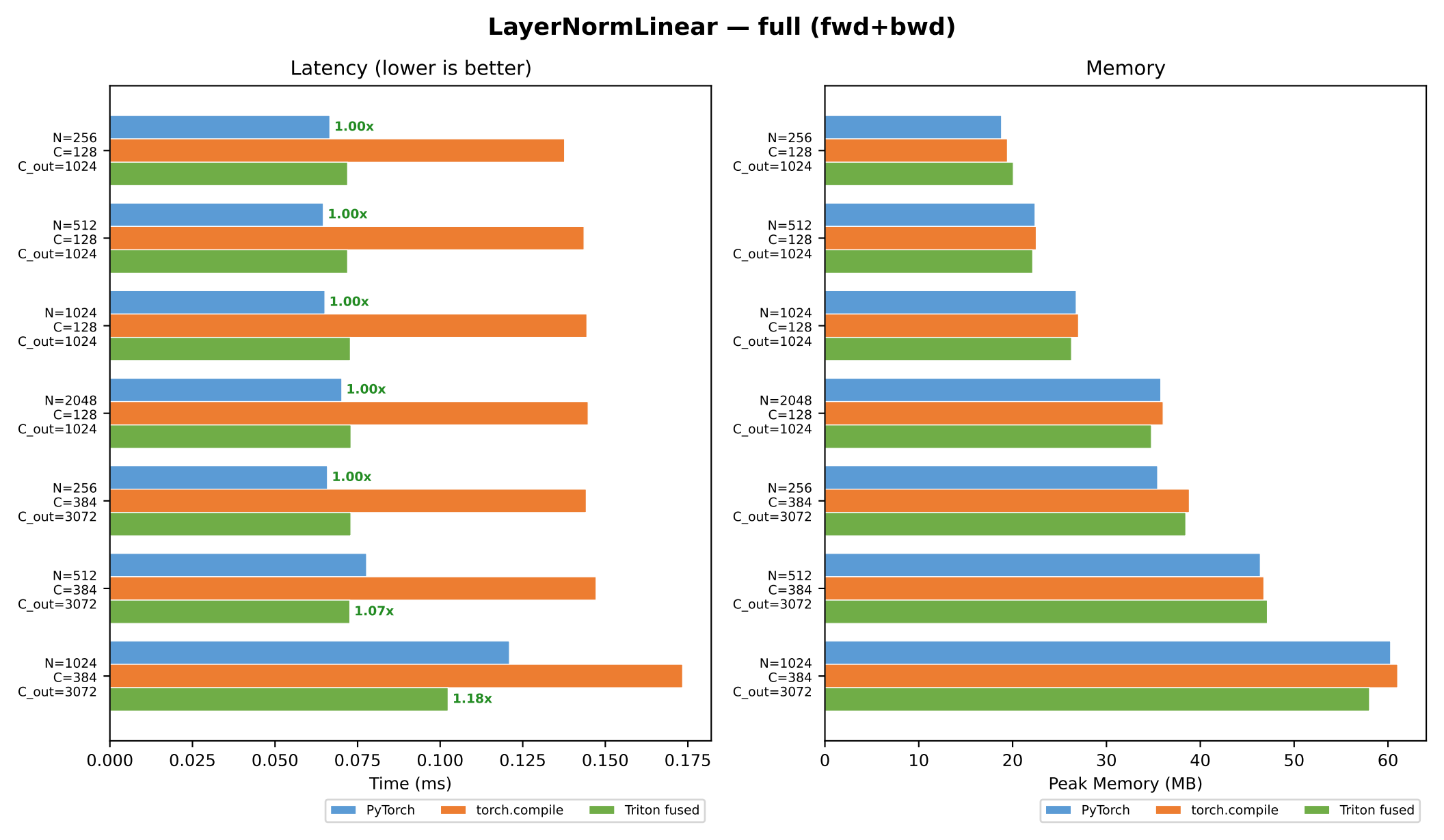

AdaTransitionBlock — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=64
N=256 C=128
N=256 C=384
N=512 C=64
N=512 C=128
N=512 C=384
N=1024 C=128
N=1024 C=384
N=2048 C=128
N=2048 C=384
1.00x
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Time (ms)
0
20
40
60
80
Peak Memory (MB)
PyTorch
torch.compile
Triton fused


GridColAttn — full (fwd+bwd)
Latency (lower is better)
N=384
C=128
H=4
2.69x
N=512
C=128
H=4
2.95x
N=640
C=128
H=4
3.17x
0
10
20
30
40
50
60
Time (ms)
PyTorch
LN+Transpose
EvoAttention
EvoAttn + PairBias
Memory
N=384
C=128
H=4
N=512
C=128
H=4
N=640
C=128
H=4
0
5000
10000
15000
20000
25000
PyTorch
LN+Transpose
EvoAttention
EvoAttn + PairBias

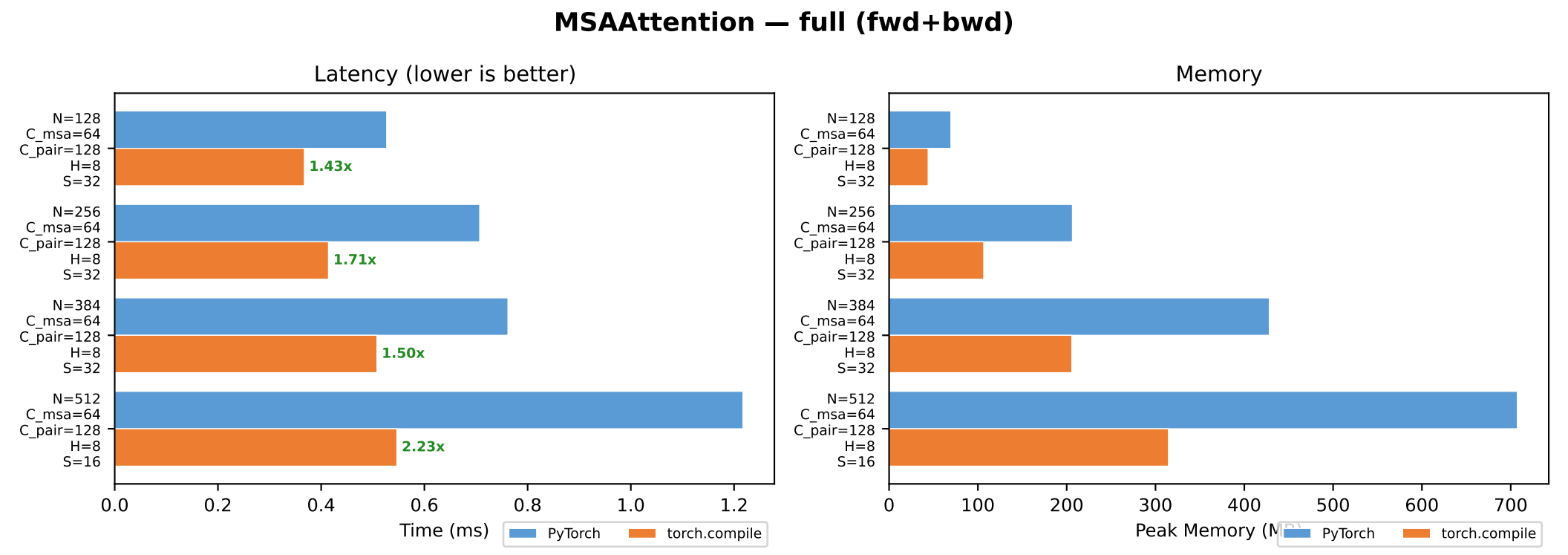

GridSelfAttention (EvoformerAttention) — full (fwd+bwd)
Latency (lower is better)
Memory
N=128
C=128
H=4
N=256
C=128
H=4
N=384
C=128
H=4
N=512
C=128
H=4
N=768
C=128
H=4
1.26x
2.64x
3.13x
3.33x
1.00x
0
5
10
15
20
25
30
35
40
Time (ms)
0
2000
4000
6000
8000
10000
12000
14000
16000
Peak Memory (MB)
PyTorch
torch.compile
EvoAttention

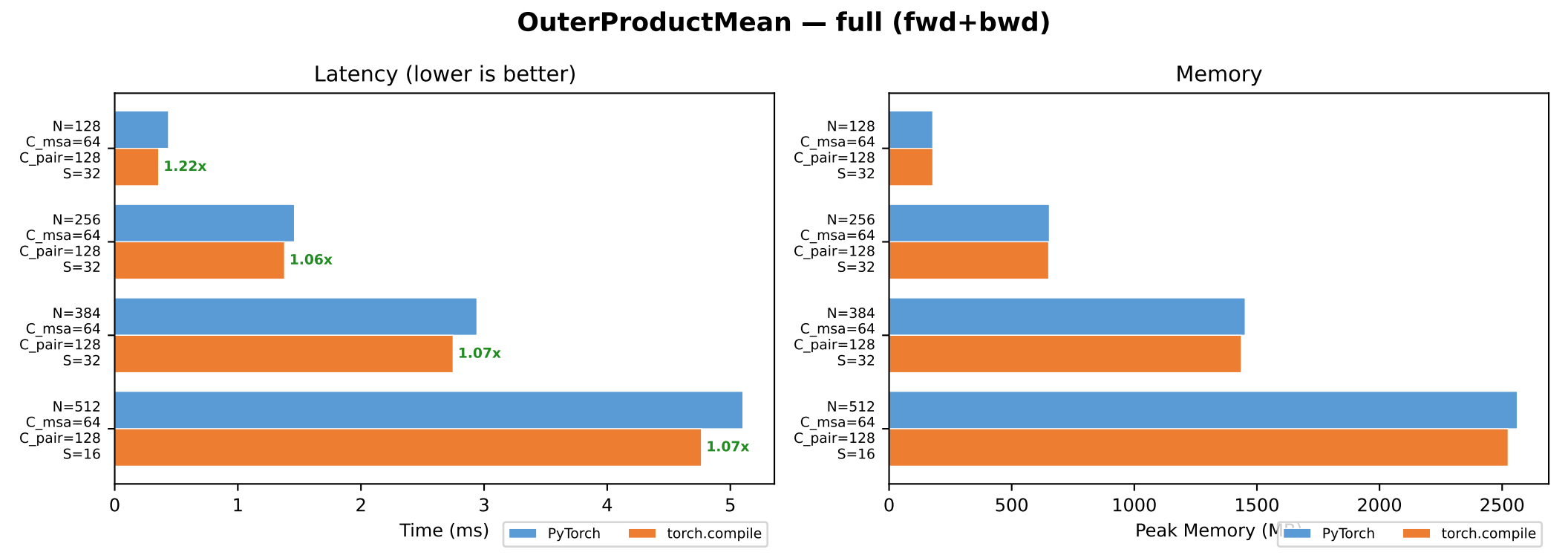

LNTranspose — full (fwd+bwd)
Latency (lower is better)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
1.00x
1.00x
1.00x
1.00x
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Time (ms)
0
500
1000
1500
2000
Peak Memory (MB)
PyTorch
Triton fused
PyTorch
Triton fused

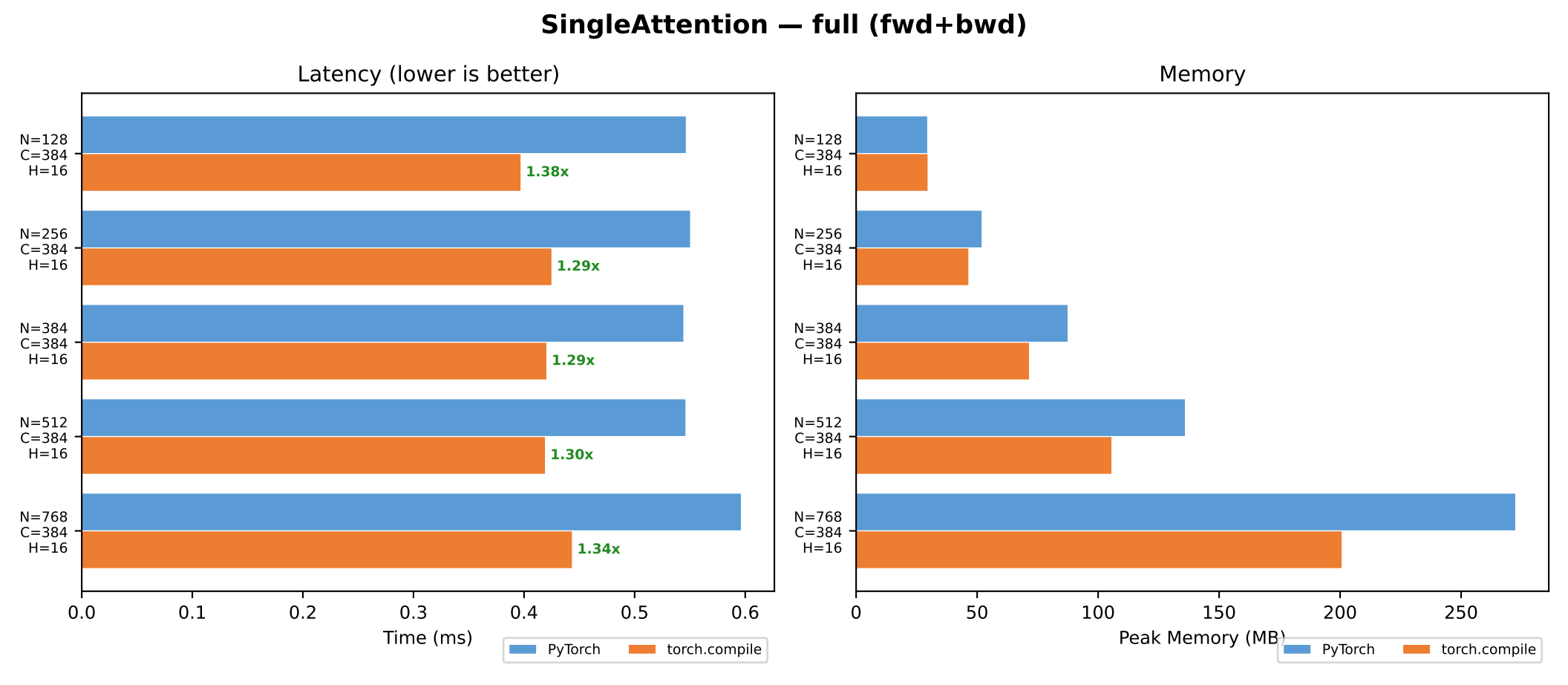

LayerNorm — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Time (ms)
0
2
4
6
8
10
12
Peak Memory (MB)
PyTorch
torch.compile

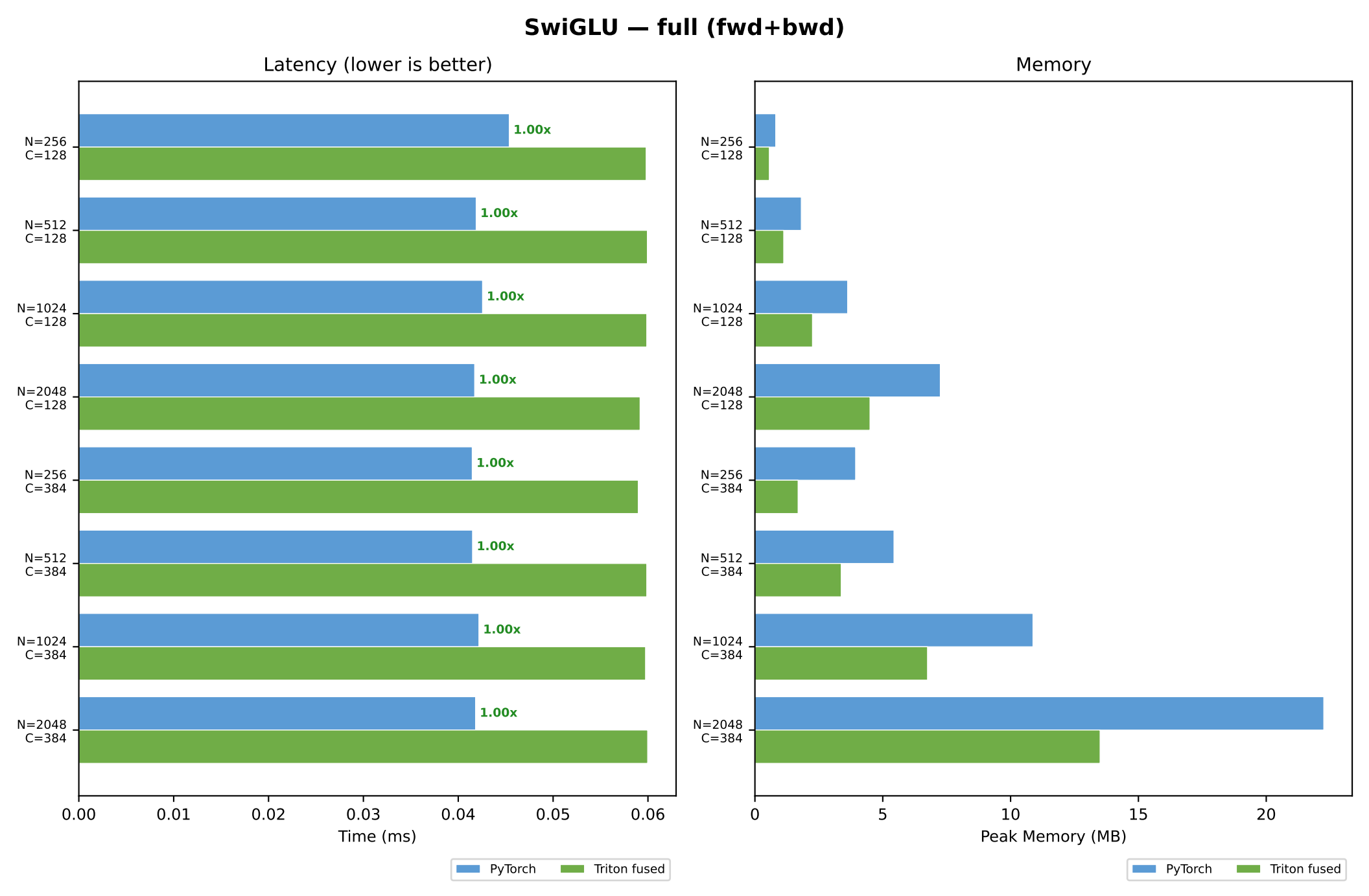

LayerNormLinear — full (fwd+bwd)
Latency (lower is better)
Memory
N=256
C=128
C_out=1024
N=512
C=128
C_out=1024
N=1024
C=128
C_out=1024
N=2048
C=128
C_out=1024
N=256
C=384
C_out=3072
N=512
C=384
C_out=3072
N=1024
C=384
C_out=3072
1.00x
1.00x
1.00x
1.00x
1.00x
1.07x
1.18x
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Time (ms)
0
10
20
30
40
50
60
Peak Memory (MB)
PyTorch
torch.compile
Triton fused

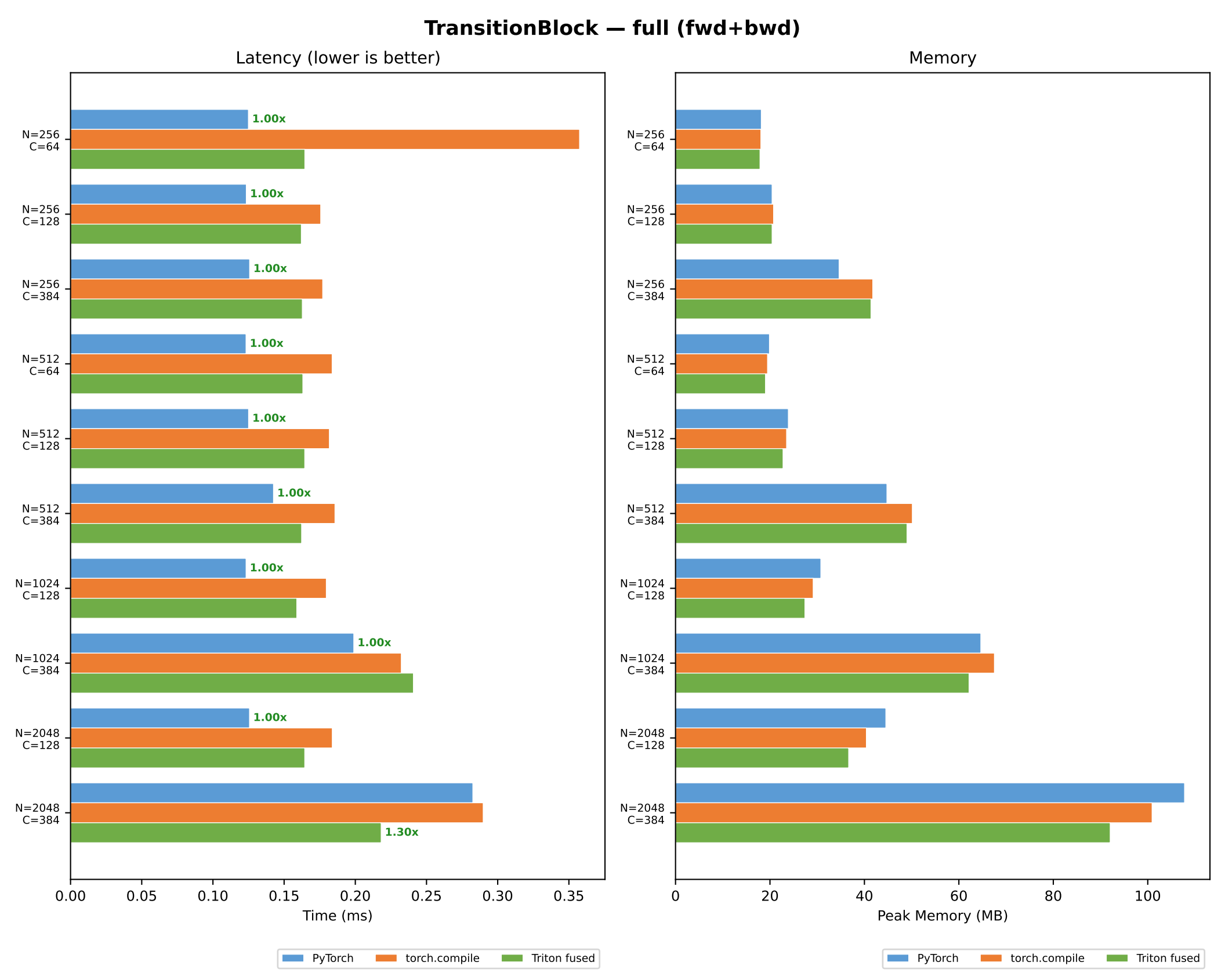

LinearNoBias — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Time (ms)
0
5
10
15
20
25
30
Peak Memory (MB)
PyTorch
torch.compile

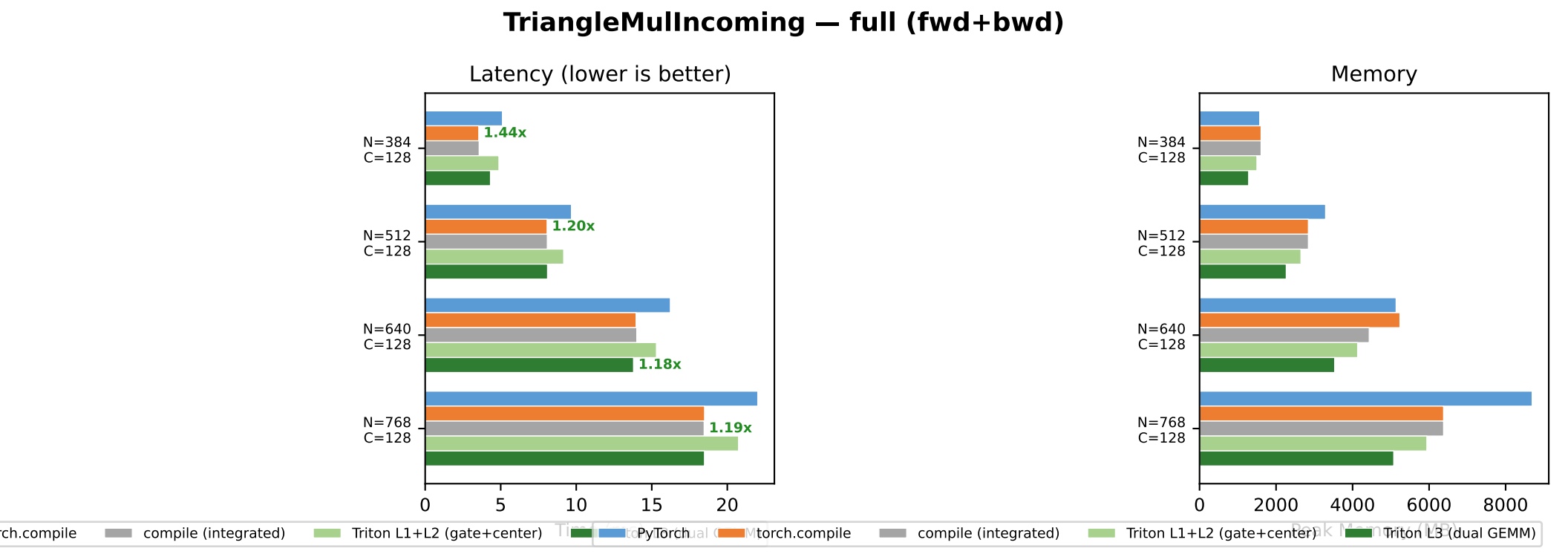

MSAAttention — full (fwd+bwd)
Latency (lower is better)
N=128
C_msa=64
C_pair=128
H=8
S=32
1.43x
N=256
C_msa=64
C_pair=128
H=8
S=32
1.71x
N=384
C_msa=64
C_pair=128
H=8
S=32
1.50x
N=512
C_msa=64
C_pair=128
H=8
S=16
2.23x
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Time (ms)
PyTorch
torch.compile
Memory
N=128
C_msa=64
C_pair=128
H=8
S=32
N=256
C_msa=64
C_pair=128
H=8
S=32
N=384
C_msa=64
C_pair=128
H=8
S=32
N=512
C_msa=64
C_pair=128
H=8
S=16
0
100
200
300
400
500
600
700
Peak Memory (M
PyTorch
torch.compile

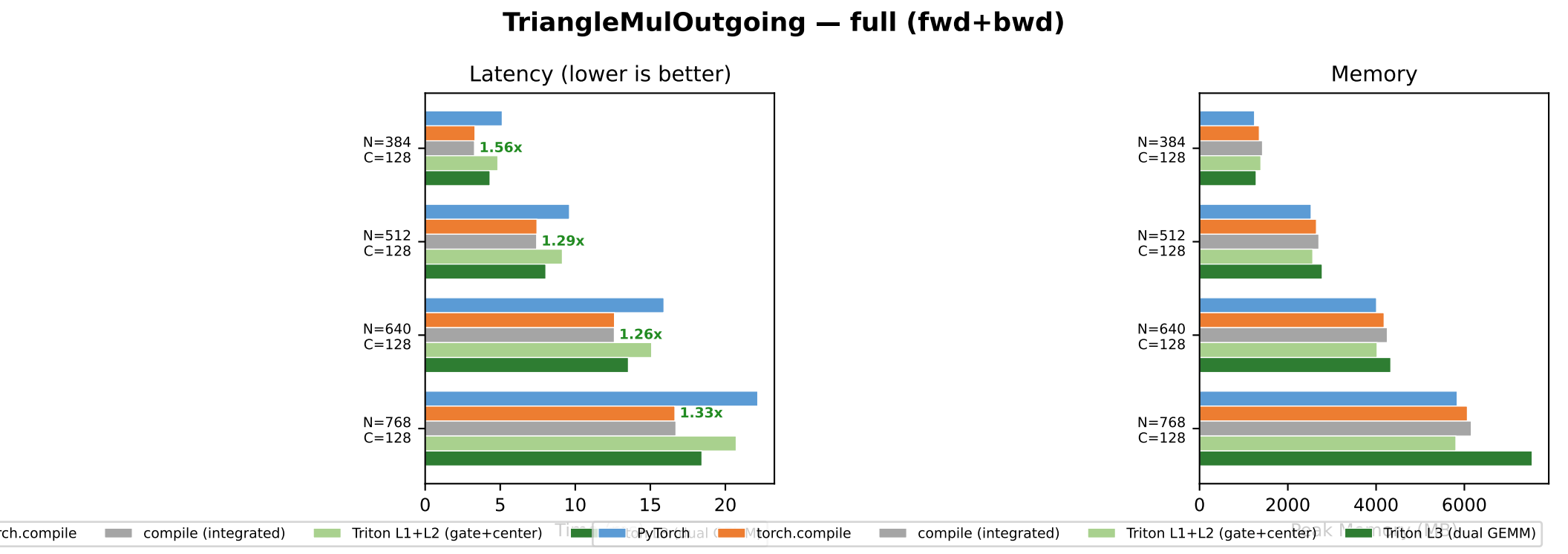

OuterProductMean — full (fwd+bwd)
Latency (lower is better)
N=128
C_msa=64
C_pair=128
S=32
1.22x
N=256
C_msa=64
C_pair=128
S=32
1.06x
N=384
C_msa=64
C_pair=128
S=32
1.07x
N=512
C_msa=64
C_pair=128
S=16
1.07x
0
1
2
3
4
5
Time (ms)
PyTorch
torch.compile
Memory
N=128
C_msa=64
C_pair=128
S=32
N=256
C_msa=64
C_pair=128
S=32
N=384
C_msa=64
C_pair=128
S=32
N=512
C_msa=64
C_pair=128
S=16
0
500
1000
1500
2000
2500
Peak Memory (M
PyTorch
torch.compile


SingleAttention — full (fwd+bwd)
Latency (lower is better)
N=128
C=384
H=16
1.38x
N=256
C=384
H=16
1.29x
N=384
C=384
H=16
1.29x
N=512
C=384
H=16
1.30x
N=768
C=384
H=16
1.34x
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Time (ms)
PyTorch
torch.compile
Memory
N=128
C=384
H=16
N=256
C=384
H=16
N=384
C=384
H=16
N=512
C=384
H=16
N=768
C=384
H=16
0
50
100
150
200
250
Peak Memory (MB)
PyTorch
torch.compile


SwiGLU — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Time (ms)
0
5
10
15
20
Peak Memory (MB)
PyTorch
Triton fused


TransitionBlock — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=64
N=256 C=128
N=256 C=384
N=512 C=64
N=512 C=128
N=512 C=384
N=1024 C=128
N=1024 C=384
N=2048 C=128
N=2048 C=384
1.00x
1.30x
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Time (ms)
0
20
40
60
80
100
Peak Memory (MB)
PyTorch
torch.compile
Triton fused


TriangleMulIncoming — full (fwd+bwd)
Latency (lower is better)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
1.44x
1.20x
1.18x
1.19x
0
5
10
15
20
0
2000
4000
6000
8000
rch.compile
compile (integrated)
Triton L1+L2 (gate+center)
PyTorch
torch.compile
compile (integrated)
Triton L1+L2 (gate+center)
Triton L3 (dual GEMM)


TriangleMulOutgoing — full (fwd+bwd)
Latency (lower is better)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
1.56x
1.29x
1.26x
1.33x
0
5
10
15
20
0
2000
4000
6000
rch.compile
compile (integrated)
Triton L1+L2 (gate+center)
PyTorch
torch.compile
compile (integrated)
Triton L1+L2 (gate+center)
Triton L3 (dual GEMM)

Recommendation Summary

Operation	Shape	Recommended	Time (ms)	Speedup
AdaTransitionBlock	N=256 / C=64	PyTorch	0.259	baseline
AdaTransitionBlock	N=256 / C=128	PyTorch	0.257	baseline
AdaTransitionBlock	N=256 / C=384	PyTorch	0.262	baseline
AdaTransitionBlock	N=512 / C=64	PyTorch	0.258	baseline
AdaTransitionBlock	N=512 / C=128	PyTorch	0.257	baseline
AdaTransitionBlock	N=512 / C=384	PyTorch	0.261	baseline
AdaTransitionBlock	N=1024 / C=128	PyTorch	0.270	baseline
AdaTransitionBlock	N=1024 / C=384	PyTorch	0.290	baseline
AdaTransitionBlock	N=2048 / C=128	PyTorch	0.271	baseline
AdaTransitionBlock	N=2048 / C=384	PyTorch	0.309	baseline
GridColAttn	N=384 / C=128 / H=4	EvoAttention	5.426	2.69x
GridColAttn	N=512 / C=128 / H=4	EvoAttn + PairBias	11.365	2.95x
GridColAttn	N=640 / C=128 / H=4	EvoAttn + PairBias	20.318	3.17x
GridSelfAttention	N=128 / C=128 / H=4	torch.compile	0.592	1.26x
GridSelfAttention	N=256 / C=128 / H=4	torch.compile	1.882	2.64x
GridSelfAttention	N=384 / C=128 / H=4	torch.compile	4.904	3.13x
GridSelfAttention	N=512 / C=128 / H=4	torch.compile	10.654	3.33x
GridSelfAttention	N=768 / C=128 / H=4	torch.compile	31.779	baseline
LNTranspose	N=384 / C=128	PyTorch	0.357	baseline
LNTranspose	N=512 / C=128	PyTorch	0.679	baseline
LNTranspose	N=640 / C=128	PyTorch	1.051	baseline
LNTranspose	N=768 / C=128	PyTorch	1.493	baseline
LayerNorm	N=256 / C=128	PyTorch	0.039	baseline
LayerNorm	N=512 / C=128	PyTorch	0.043	baseline
LayerNorm	N=1024 / C=128	PyTorch	0.042	baseline
LayerNorm	N=2048 / C=128	PyTorch	0.042	baseline
LayerNorm	N=256 / C=384	PyTorch	0.042	baseline
LayerNorm	N=512 / C=384	PyTorch	0.041	baseline
LayerNorm	N=1024 / C=384	PyTorch	0.042	baseline
LayerNorm	N=2048 / C=384	PyTorch	0.042	baseline
LayerNormLinear	N=256 / C=128 / C_out=1024	PyTorch	0.067	baseline
LayerNormLinear	N=512 / C=128 / C_out=1024	PyTorch	0.065	baseline
LayerNormLinear	N=1024 / C=128 / C_out=1024	PyTorch	0.065	baseline
LayerNormLinear	N=2048 / C=128 / C_out=1024	PyTorch	0.070	baseline
LayerNormLinear	N=256 / C=384 / C_out=3072	PyTorch	0.066	baseline

Operation	Shape	Recommended	Time (ms)	Speedup
LayerNormLinear	N=512 / C=384 / C_out=3072	Triton fused	0.073	1.07x
LayerNormLinear	N=1024 / C=384 / C_out=3072	Triton fused	0.102	1.18x
LinearNoBias	N=256 / C=128	PyTorch	0.040	baseline
LinearNoBias	N=512 / C=128	PyTorch	0.041	baseline
LinearNoBias	N=1024 / C=128	PyTorch	0.042	baseline
LinearNoBias	N=2048 / C=128	PyTorch	0.042	baseline
LinearNoBias	N=256 / C=384	PyTorch	0.045	baseline
LinearNoBias	N=512 / C=384	PyTorch	0.042	baseline
LinearNoBias	N=1024 / C=384	PyTorch	0.063	baseline
LinearNoBias	N=2048 / C=384	PyTorch	0.044	baseline
MSAAttention	N=128 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.367	1.43x
MSAAttention	N=256 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.414	1.71x
MSAAttention	N=384 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.508	1.50x
MSAAttention	N=512 / C_msa=64 / C_pair=128 / H=8 / S=16	torch.compile	0.547	2.23x
OuterProductMean	N=128 / C_msa=64 / C_pair=128 / S=32	torch.compile	0.357	1.22x
OuterProductMean	N=256 / C_msa=64 / C_pair=128 / S=32	torch.compile	1.380	1.06x
OuterProductMean	N=384 / C_msa=64 / C_pair=128 / S=32	torch.compile	2.749	1.07x
OuterProductMean	N=512 / C_msa=64 / C_pair=128 / S=16	torch.compile	4.766	1.07x
SingleAttention	N=128 / C=384 / H=16	torch.compile	0.397	1.38x
SingleAttention	N=256 / C=384 / H=16	torch.compile	0.425	1.29x
SingleAttention	N=384 / C=384 / H=16	torch.compile	0.421	1.29x
SingleAttention	N=512 / C=384 / H=16	torch.compile	0.419	1.30x
SingleAttention	N=768 / C=384 / H=16	torch.compile	0.444	1.34x
SwiGLU	N=256 / C=128	PyTorch	0.045	baseline
SwiGLU	N=512 / C=128	PyTorch	0.042	baseline
SwiGLU	N=1024 / C=128	PyTorch	0.043	baseline
SwiGLU	N=2048 / C=128	PyTorch	0.042	baseline
SwiGLU	N=256 / C=384	PyTorch	0.041	baseline
SwiGLU	N=512 / C=384	PyTorch	0.042	baseline
SwiGLU	N=1024 / C=384	PyTorch	0.042	baseline
SwiGLU	N=2048 / C=384	PyTorch	0.042	baseline
TransitionBlock	N=256 / C=64	PyTorch	0.125	baseline
TransitionBlock	N=256 / C=128	PyTorch	0.123	baseline
TransitionBlock	N=256 / C=384	PyTorch	0.126	baseline
TransitionBlock	N=512 / C=64	PyTorch	0.123	baseline

Operation	Shape	Recommended	Time (ms)	Speedup
TransitionBlock	N=512 / C=128	PyTorch	0.125	baseline
TransitionBlock	N=512 / C=384	PyTorch	0.143	baseline
TransitionBlock	N=1024 / C=128	PyTorch	0.123	baseline
TransitionBlock	N=1024 / C=384	PyTorch	0.199	baseline
TransitionBlock	N=2048 / C=128	PyTorch	0.126	baseline
TransitionBlock	N=2048 / C=384	Triton fused	0.218	1.30x
TriangleMulIncoming	N=384 / C=128	torch.compile	3.546	1.44x
TriangleMulIncoming	N=512 / C=128	torch.compile	8.069	1.20x
TriangleMulIncoming	N=640 / C=128	Triton L3 (dual GEMM)	13.791	1.18x
TriangleMulIncoming	N=768 / C=128	compile (integrated)	18.465	1.19x
TriangleMulOutgoing	N=384 / C=128	compile (integrated)	3.283	1.56x
TriangleMulOutgoing	N=512 / C=128	compile (integrated)	7.425	1.29x
TriangleMulOutgoing	N=640 / C=128	compile (integrated)	12.602	1.26x
TriangleMulOutgoing	N=768 / C=128	torch.compile	16.637	1.33x